An uninterruptible power supply is called UPS. It is a device that permits supply to keep on running for a short period of time, when the primary power is off.

UPS contains a battery that "kicks in" when the device senses a loss of power from the primary source (as in fig. 4.11).

If you are using computer, when the UPS notifies you of the power loss, you have time to save and data you are working on and exit, before the secondary power source runs out. When all power runs out, any data in computers Random Access Memory (RAM) is erased when power surges occur, a UPS intercepts the surge, so that it doesn't damage the computer.

4.8 MAINTENANCE OF BATTERIES

How does UPS work?

In a continuous UPS, the computer is always running short of battery power and the battery is continuously being recharged. The battery charger continuously produces DC power, which the inverter continuously turns back into 120 volt AC power. If the power fails, the battery provides power to the inverter.

Components of UPS (Ref. fig. 4.12)

1. The Static Bypass
2. The Rectifier
3. The Battery
4. The Inverter

Types of UPS

Range		Types
0.5 to 3 KVA	-	Line interactive
0.5 to 5 KVA	-	Stand by Online Hybrid
3.0 to 15 KVA	-	Stand by Ferro Double
5.0 to 5000KVA	-	Conversion Online

1. Battery should be cleaned properly
2. Cable connection of the battery needs to be clean and tightened, Many battery problems are caused by dirty and loose connection.
3. The fluid level of the battery will always be higher at a full charge.
4. Distilled water alone is the best for filling because other types of water are loaded with chemicals and minerals that are harmful to the battery. Don't over fill the battery especially in warm weather.
5. Use silicon seals in the cable leads. Coat the cable washer end with grease or petroleum jelly (vaseline).

4.9 DO'S AND DON'T OF STORAGE BATTERY

Do's

1. Store batteries in a clean, ventilated and dry area.
2. Store batteries in a fully charged state.
3. Ensure the correct polarity connection when recharging.
4. Follow proper recharging schedules to prevent overcharging.
5. Keep the battery away from spark, heat and sources of fire.
6. Use proper size of cables along with correct plugs.
7. Charge the batteries immediately after it is discharged.

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